

Comprehensive Report: Fundamentals of Generative AI and Large Language Models

Name: HARINI S K

Register no: 212225230092

Slot: T2-H16

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ENGINEERING

1.1 Subject: Generative AI, LLMs, and 2024–2025 AI Tool Landscape

1.2 Target Audience: Students and Technical Professionals

1.3 Table of Contents

1. Experiment:

Develop a comprehensive report for the following exercises:

1 Explain the foundational concepts of Generative AI.

. Focusing on Generative AI architectures. (like transformers).

2 Generative AI architecture and its applications.

. Generative AI impact of scaling in LLMs.

3 Explain about LLM and how it is built.

4 Foundational Concepts of Generative AI

. Generative AI is a branch of Artificial Intelligence that creates new content such as text, images, music, videos, and code by learning patterns from existing data. Instead of only analyzing data, it can generate original outputs that resemble human-created work.

1. Data and Patterns

Generative AI models are trained using huge amounts of data. The system studies patterns, structures, grammar, colors, sounds, and relationships in the data.

Example: A text model learns sentence structures from books and websites.

2. Model Architecture

Generative AI uses advanced neural network models to process and learn information.

Common architectures include:

1. Transformer Model
2. Generative Adversarial Network (GANs)
3. Variational Autoencoder (VAEs)
4. Diffusion Model

3. Generation Process

After training, the AI can generate new content based on learned patterns.

The process works like this:

1. User gives an input or prompt.
2. The model predicts the most suitable output.
3. AI generates new content step by step.

Examples:

- Writing essays or stories □
- Generating music
- 4. Control and Conditioning

Users can guide AI outputs through prompts or instructions.

This is called conditioning.

Example prompts:

- “Write a poem about nature.”
- “Create a futuristic city image.”

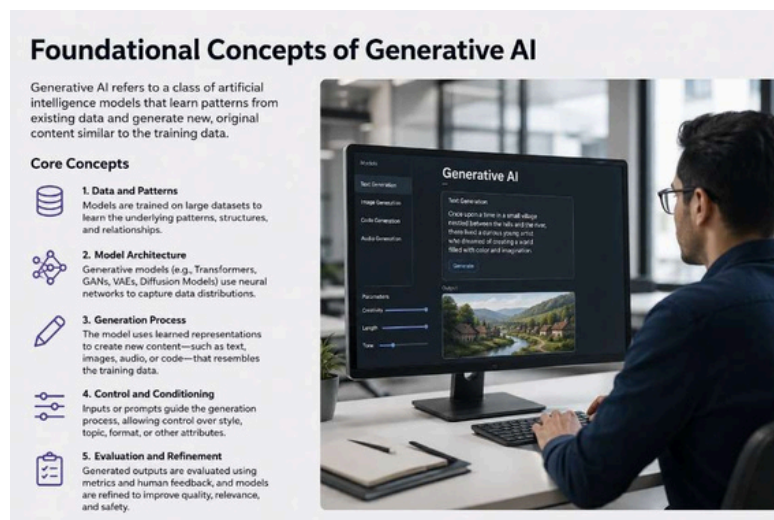
The prompt controls:

- Style
- Tone
- Topic

5. Evaluation and Refinement

Generated content is checked for:

- Accuracy
- Quality



Focusing on Generative AI architectures. (like transformers).

1. Transformer Architecture

Transformer Model is the most popular architecture used in modern AI systems like chatbots and text generators.

Example

Input:

“Artificial Intelligence is...”

Output prediction:

“changing the world.”

2. GAN (Generative Adversarial Network)

Generative Adversarial Network is mainly used for creating realistic images.

Example

- AI-generated human faces

- Deepfake videos

3. VAE (Variational Autoencoder)

Variational Autoencoder is used for learning compressed data patterns and recreating them.

Example

- Image reconstruction

- Noise removal

4. Diffusion Model

Diffusion Model is widely used in modern image-generation systems.

Example

- AI art generation

- Text-to-image systems

5. Autoregressive Models

These models generate content one step at a time.

Example

- Text generation

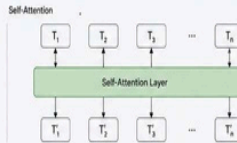
- Music generation

Generative AI Architectures

The foundational models behind modern generative AI

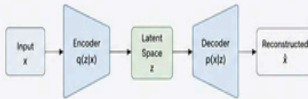
1. Transformer (e.g., GPT)

- Based on self-attention mechanism
- Excels in text generation, translation, summarization, and reasoning
- Processes sequential data in parallel, capturing long-range dependencies



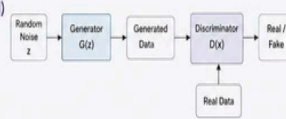
2. Variational Autoencoder (VAE)

- Learns a compressed latent representation of data
- Encoder maps input to a latent space; decoder reconstructs the output
- Used for generation, data compression, and anomaly detection



3. Generative Adversarial Network (GAN)

- Composed of Generator and Discriminator networks
- Generator creates fake data; discriminator tries to distinguish real vs. fake
- Produces highly realistic outputs



4. Diffusion Model (e.g., DALL-E 3, Stable Diffusion)

- Adds noise to data gradually and learns to reverse the process
- Generates high-quality images and data
- State-of-the-art for image synthesis



5. Autoregressive Models

- Generate data step-by-step, conditioning on previous outputs
- Common in text, code, and music generation

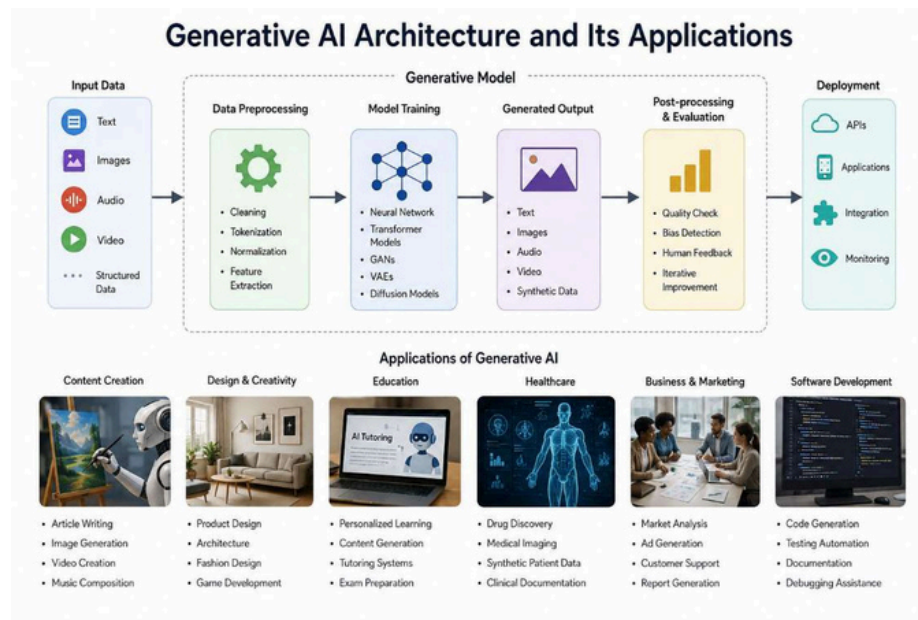


Summary

Different architectures power generative AI systems. Transformers lead in text, GANs in realistic image synthesis, VAEs in latent representation learning, and diffusion models in high-quality generation.

Generave AI architecture and its applicaons.

- Architecture of Generave AI
 1. Input Data
 2. Data Processing
 3. Model Training
 4. Content Generaon
 5. Output Evaluaon
- Applicaoons of Generave AI
 1. Content Crea on
 2. Educa on
 3. Healthcare
 4. Business
 5. Soware Development



Impact of Scaling in LLMs – Short Note

Scaling in Large Language Models (LLMs) means increasing the model size, training data, and computing power.

*Impact of Scaling

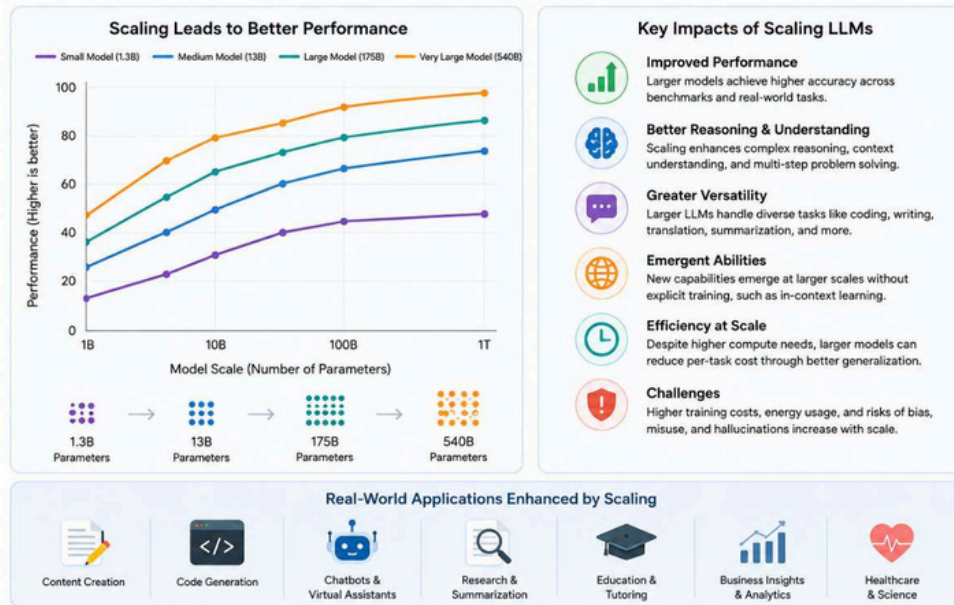
1. Improves accuracy and understanding
2. Generates better text, images, and code
3. Supports multiple languages
4. Enhances reasoning and problem-solving
5. Increases creativity and performance

Challenges

1. High cost and energy usage
2. Requires powerful computers
3. May produce biased or incorrect outputs

Generative AI: Impact of Scaling in LLMs

Larger models trained on more data and compute show significant improvements in capability, accuracy, and versatility.



Large Language Model (LLM) – Simple Explanaon

A Large Language Model (LLM) is a type of Artificial Intelligence that understands and generates human language.

Examples:

- ChatGPT
- Gemini
- Claude

LLMs can:

- Answer quesons
- Write text

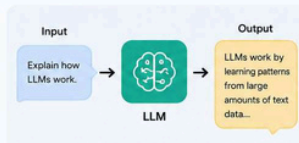
How is an LLM Built?

1. Data Collecon

2. Data Processing
3. Model Training
4. Fine-Tuning
5. Deployment

What is an LLM?

A Large Language Model (LLM) is an AI model trained on huge amounts of text data to understand and generate human-like text. Examples: GPT, Llama, PaLM, Claude, Gemini.



How is an LLM Built?

Building an LLM involves several key steps:

- 1 **Data Collection**
Collect massive amounts of text from books, websites, articles, code, and more.
- 2 **Data Preprocessing**
Clean the data, remove duplicates, fix errors, and convert it into tokens (words/pieces).
- 3 **Model Architecture**
Use a neural network architecture called Transformer, which is designed to understand relationships in text.
- 4 **Training (Pre-training)**
Train the model on the prepared data to predict the next word in a sequence. This helps the model learn patterns, grammar, facts, and reasoning.

LLM Building Process

